THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 08

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The distance between $A(1, 2)$ and $B(4, 6)$ is: 1

(a) 5 units **(b)** 7 units **(c)** 6 units **(d)** 4 units

2. Which of the following is a linear polynomial? 1

(a) $5x - 8$ **(b)** $x^3 - x$ **(c)** 7 **(d)** $x^2 + 1$

3. The set of all rational numbers is denoted by: 1

(a) $\mathbb{R}$ **(b)** $\mathbb{Z}$ **(c)** $\mathbb{Q}$ **(d)** $\mathbb{N}$

4. Expand $(3x - 4)^2$: 1

(a) $9x^2 - 24x + 16$ **(b)** $9x^2 - 16$ **(c)** $9x^2 + 24x + 16$ **(d)** $9x^2 - 12x + 16$

5. In a circle of radius 13 cm, a chord is 12 cm from the centre. The length of the chord is: 1

(a) 8 cm **(b)** 10 cm **(c)** 12 cm **(d)** 5 cm

6. The perpendicular bisector of a chord of a circle always passes through: 1

(a) the centre of the circle **(b)** no fixed point

(c) the midpoint of the arc only **(d)** a point outside the circle

7. A median of a triangle divides it into two triangles of: 1

(a) equal perimeter **(b)** equal angles

(c) equal area **(d)** one double the other

8. The number of outcomes in the sample space when a coin is tossed and a die is rolled is: 1

(a) 6 **(b)** 24 **(c)** 14 **(d)** 12

9. For the sequence $t_n = n^2 - 2n + 3$, the value of t_7 is: 1

(a) 76 **(b)** 38 **(c)** 39 **(d)** 37

10. The number 48 has: 1

(a) 11 factors; it is a perfect square **(b)** 10 factors; it is not a perfect square

(c) 9 factors; it is a perfect square **(d)** 12 factors; it is not a perfect square

11. "If a number is divisible by both 3 and 5, then it is divisible by 15." Its converse "If a number is divisible by 15, then it is divisible by both 3 and 5" is: 1

(a) true **(b)** false, since 15 is odd

(c) false, since 30 is divisible by 15 **(d)** neither true nor false

12. 80 mL of a solution with 25% salt is mixed with 120 mL of a solution with 15% salt. The concentration of salt in the mixture is: 1

(a) 19% **(b)** 20% **(c)** 18% **(d)** 21%

13. **Assertion (A):** In a 100% stacked bar chart, a longer segment for Bakery A than for Bakery B in the bread category means that A sold more bread than B. 1

Reason (R): A 100% stacked bar shows the percentage share within each bakery's own sales, not the actual amounts.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

14. Using the long-division (Āryabhaṭa) form of Euclid's algorithm, the number of reductions needed for gcd(825, 375) is: 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 3

15. While adding 2764 and 3589 by the column algorithm, the number of columns that produce a carry of 1 is: 1

(a) 2 **(b)** 5 **(c)** 4 **(d)** 3

16. In $\triangle ABC$, P and Q are the midpoints of AB and AC and BC = 14 cm. The length of PQ is: 1

(a) 14 cm **(b)** 9 cm **(c)** 7 cm **(d)** 28 cm

17. In a parallelogram ABCD, $\angle A = 48°$. Then $\angle B$ is: 1

(a) 66° **(b)** 312° **(c)** 132° **(d)** 48°

18. The line ax + by + c = 0 always passes through the origin when: 1

(a) a = b **(b)** c = 0 **(c)** a = 0 **(d)** b = 0

19. Taking $\pi = \frac{22}{7}$, the volume of a cylinder of radius 14 cm and height 3 cm is: 1

(a) 924 cm^3 **(b)** 1848 cm^3 **(c)** 3696 cm^3 **(d)** 264 cm^3

20. The volume of a cube is 125 cm^3. Its total surface area is: 1

(a) 100 cm^2 **(b)** 150 cm^2 **(c)** 75 cm^2 **(d)** 25 cm^2

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Show that the identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ holds for $a = 2,\ b = 1$. 2

22. Points C and D lie on the same side of AB and $\angle ACB = \angle ADB$. What can you say about A, B, C, D? 2

23. In $\triangle ABC$ of area 48 cm^2, AD is a median. Find the area of $\triangle ABD$ and give the reason. 2

24. Find the 10th and 26th terms of the AP $3, 8, 13, 18, \ldots$ 2

25. A bar in a 100% stacked chart shows 25% for lighting. Total consumption is 800 units. How many units are for lighting? 2

26. What is 17 mod 5 and 33 mod 7? What do they represent? 2

27. The diagonals of parallelogram ABCD meet at O, with AC $= 14$ cm and BD $= 10$ cm. Find OA and OB. 2

28. Without solving, decide whether $3x + 2y = 5$ and $6x + 4y = 11$ have a solution. Give a reason. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 40–60 words each.

29. The points $(-3, 6)$, $(-2, 4)$, $(0, 0)$, $(1, -2)$ lie on a line. Find its equation and its slope. What does the slope tell? 3

30. A shop earned ₹8000 from books at 30% profit and ₹1000 from covers at 50% profit. Find the overall profit percentage. 3

31. Give two examples (one from number theory and one from geometry) of propositions that are true but whose converses are false. 3

32. Which term of the GP $2, 6, 18, \ldots$ is 486? 3

33. Show that the point $(0, -1)$ is equidistant from $(-4, 3)$ and $(4, 3)$. Hence name the type of triangle formed by the three points, with reason. 3

34. A bag of 12 balls has red ones with $P(\text{red}) = \frac{1}{3}$. 3
(i) How many red balls?
(ii) If 2 more red balls are added, find the new $P(\text{red})$.

35. Is $(a + b - c)^2 + (a - b + c)^2 + (a - b - c)^2 = 2a^2 + 2b^2 + 2c^2$ an identity? Test with $a = 1, b = 1, c = 1$. 3

OR

Use $(x - y)(x^2 + xy + y^2) = x^3 - y^3$ to factorise $27a^3 - 8b^3$. 3

36. Show that a cyclic parallelogram is a rectangle. 3

OR

Prove that equal chords of a circle subtend equal angles at the centre. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** The chapter says the number of reductions in Euclid's subtraction algorithm for gcd(2k + 1, 2) is about k. 5
(i) For k = 5, write the pair-chain for gcd(11, 2) and count the reductions.
(ii) Write the division-algorithm chain.
(iii) Compare.
(iv) What does this show about the efficiency of the two algorithms?

OR

Write an algorithm prime(n). Execute it for n = 21 and n = 23. 5

38. A cone, a sphere and a cylinder each have radius r; the cone and the cylinder have height $2r$ (the sphere's diameter). Show that their volumes are in the ratio $1 : 2 : 3$. Verify with $r = 3$ cm (leave answers in terms of π). 5

OR

The radius of a cylinder is increased by 20% while its height is unchanged. Find the percentage change in
(i) the curved surface area,
(ii) the volume, and
(iii) the total surface area when $h = r$. 5

39. Find the values of $(i)\ 1104^2$ and $(ii)\ 198^2$ using suitable identities. 5
(iii) Which identity makes each easier?

OR

Case study: Explore $(x - y)(x^2 + xy + y^2)$ and $(x + y)(x^2 - xy + y^2)$. 5
(i) Multiply and state the identities.
(ii) Factorise $27p^3 + 8q^3$ and $64 - a^3$.

40. (i) Plot the points (n, t_n) for the AP $1, 5, 9, 13, 17$ and say what you observe. 5
(ii) Plot the points for the GP $3, 6, 12, 24, 48$ and compare.
(iii) What does this tell you about growth in an AP and a GP?

OR

(i) Prove that $t_n + t_{n-1} = n^2$ for triangular numbers. 5
(ii) Verify for $n = 10$.
(iii) Find the 17th and 80th triangular numbers.

END OF QUESTION PAPER **Total: 80 Marks**

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